

Problem 1.

- a) See solution document.
- b) $a = \frac{F}{m} \cos \theta$
- c) $n = mg - F \sin \theta$

Problem 2.

- a) $a = \frac{F}{m_A + m_B} - g$
- b) $S = \frac{m_B}{m_A + m_B} F$
- c) $a = -3.82 \text{ m/s}^2$

Problem 3.

- a) $\sum F = 4.0 \text{ N}$
- b) $[2.0 \text{ s}, 6.0 \text{ s}]$.
- c) $\sum F = 3.0 \text{ N}$

Problem 4.

F.

Problem 5.

- a) See solution document.
- b) $a = \frac{F}{m} \cos \theta - g \sin \theta$
- c) $n = F \sin \theta + mg \cos \theta$

Problem 6.

- a) $a = \frac{F}{m_A + m_B + m_C}$
- b) $S_{BC} = \frac{m_C}{m_A + m_B + m_C} F$
- c) $S_{AB} = \frac{m_B + m_C}{m_A + m_B + m_C} F$

Problem 7.

- a) See solution document.
- b) See solution document.